

Dissecting the mechanisms of low-intensity focused ultrasound neuromodulation

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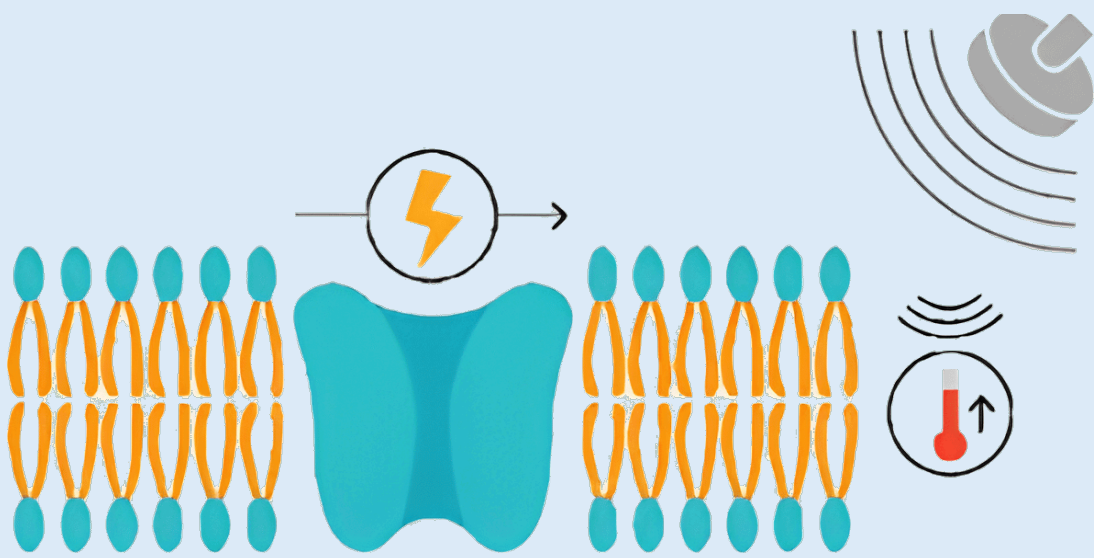
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Low-intensity focused ultrasound offers a non-invasive route to modulate neural activity with high spatial selectivity, but its cellular mechanisms remain unresolved and parameter-dependent. The **SEVERUS** project addresses this gap by testing alternative, non-exclusive hypotheses for focused ultrasound stimulation in neuronal networks.



IN SILICO: MULTISCALE MODELLING

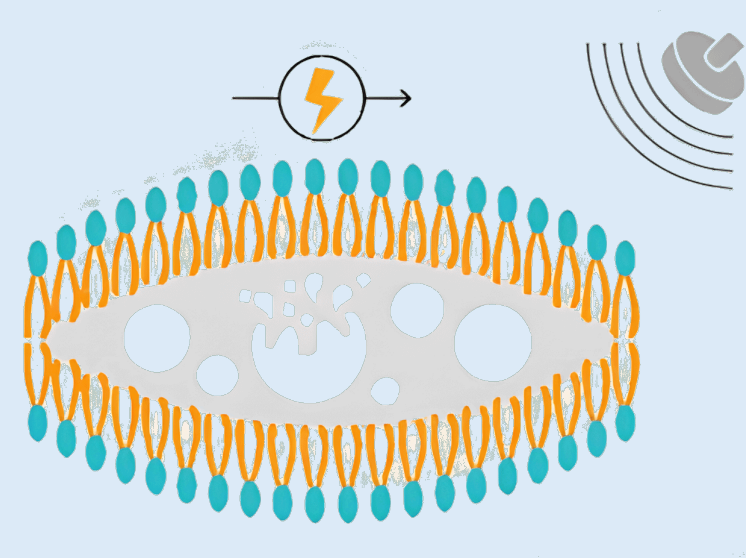
Proposed neuromodulation mechanisms



(i) Thermal transients

Heating shifts conductance and channel kinetics, short of damage.

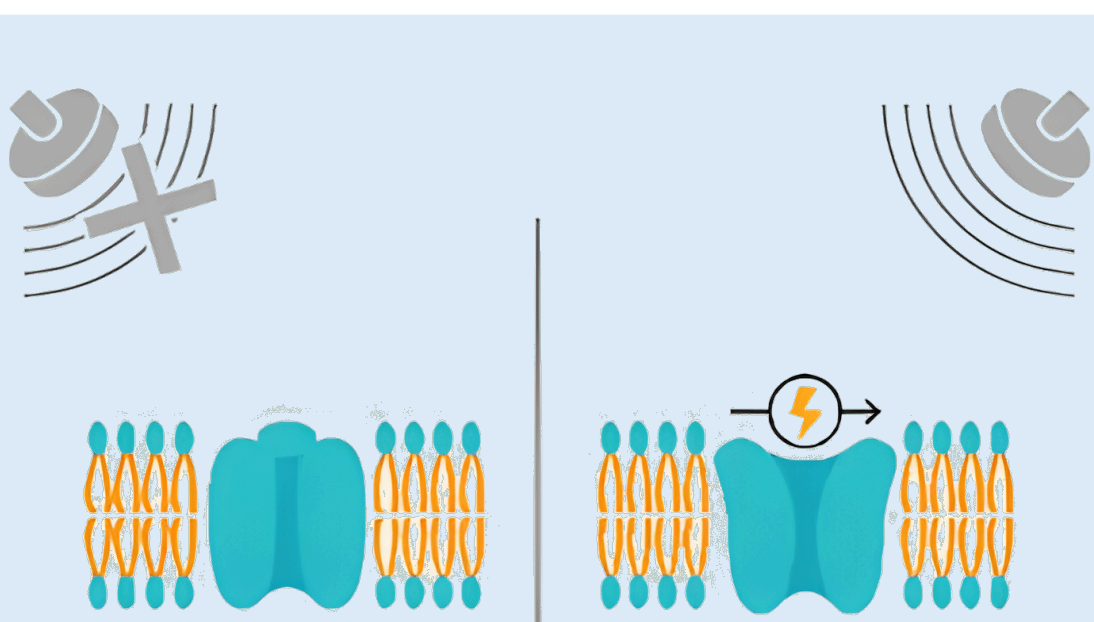
PREDICTS time-averaged intensity, not peak pressure.



(ii) Intramembrane cavitation

Nanobubbles in the bilayer change capacitance; displacement currents follow.

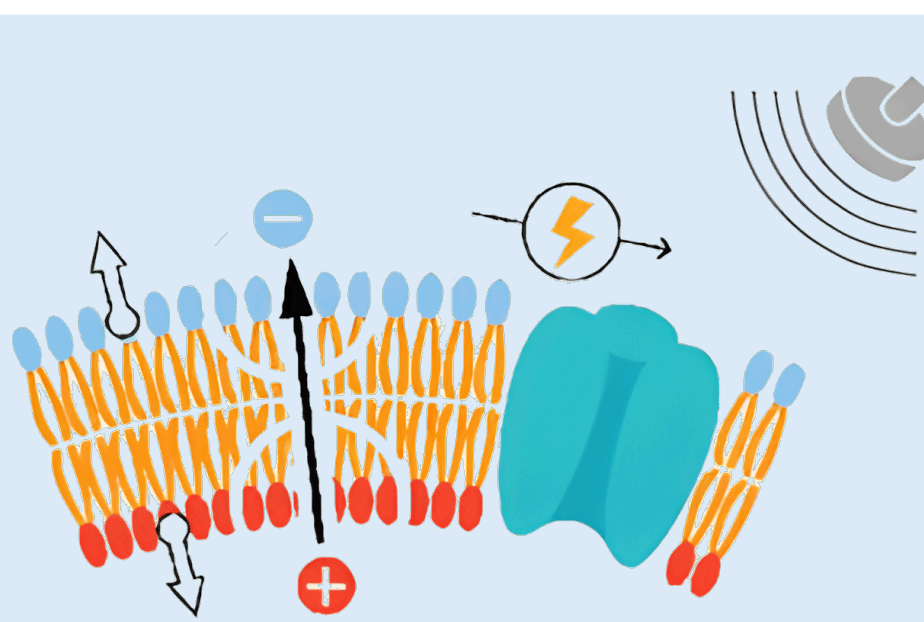
PREDICTS a frequency-dependent pressure threshold, indifferent to degassing.



(iii) Mechanosensitive gating

Force and deformation open Piezo- and TRP-like channels.

PREDICTS sensitivity to channel blockers; cell-type specificity.



(iv) Membrane biophysics

Thickness, curvature and lipid state move capacitance. The panel shows flexoelectricity.

PREDICTS an effect that survives blockade of (iii).

(v) Streaming, shear and radiation force

Force on the cells and on the matrix they are coupled to; streaming adds shear.

PREDICTS a slower time course. Gel embedding separates the two.

(vi) Glial relay

Ultrasound acts on astrocytes through TRPA1; gliotransmitter release onto NMDARs does the rest. (Oh 2019)

PREDICTS nothing in neuron-only culture.

Sweep frequency, intensity, duty cycle and pulse duration, *in silico* and verify experimentally *in vitro*.

Modelling stack

TOOL	WHAT IT COMPUTES	OUTPUT
CAD	Chip geometry: channels, gel compartment, acoustic windows.	→ meshed stack
COMSOL	Acoustic and thermal field in that geometry, not in free water.	→ pressure and temperature at the construct
nest::ml	Ion-channel kinetics and the neuron models they define.	→ generated model code
nest::	Spiking and network synchrony across the construct.	→ predicted spiking
	Coordinates and couples stages: field into membrane mechanics and capacitance, and that into the transmembrane current the channel models take.	

IN VITRO: INNOVATIVE 3D PLATFORM

A history of confounds

Standing waves. Rigid bottoms reflect, high sensitivity. (Hensel 2011; Secomski 2017; Tretbar 2025)

Exposure uncertainty. Up to ~700% across standard well plates. (Leskinen & Hynynen 2012)

Transducer heating. Above roughly 5% duty cycle, dry-coupled. (Tretbar 2025)

Readout motion. ~40% of widefield calcium variance during sonication. (Kitahara & Tateno 2026)

Absorber on the return path, 40° tilt.

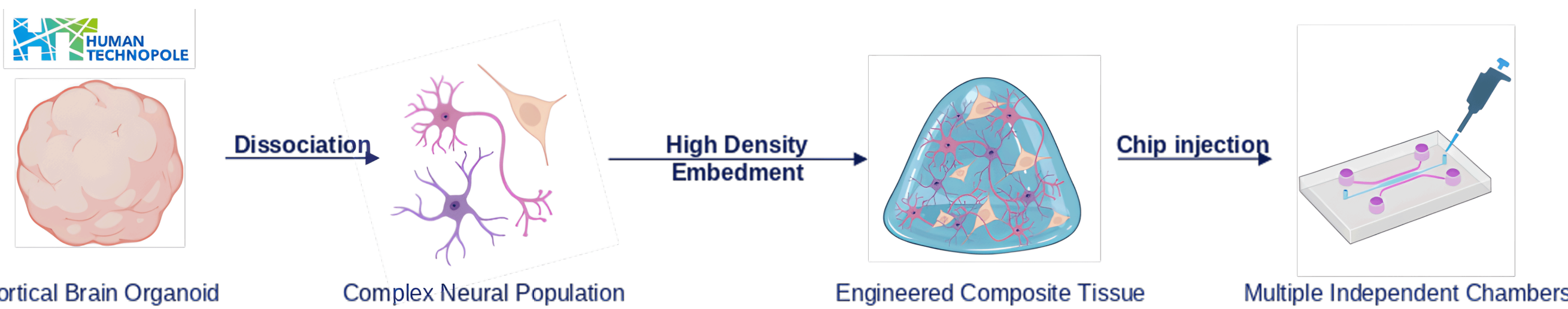
Field reported at the cell plane: tank scan plus accurate model of real stack.

Thermal readout in the rig; duty cycle set against a measured ceiling.

Cells held in gel.

3D organ-on-chip platform

A perfused, compartmentalised microfluidic platform hosting a 3D neural construct under characterised sub-MHz exposure.



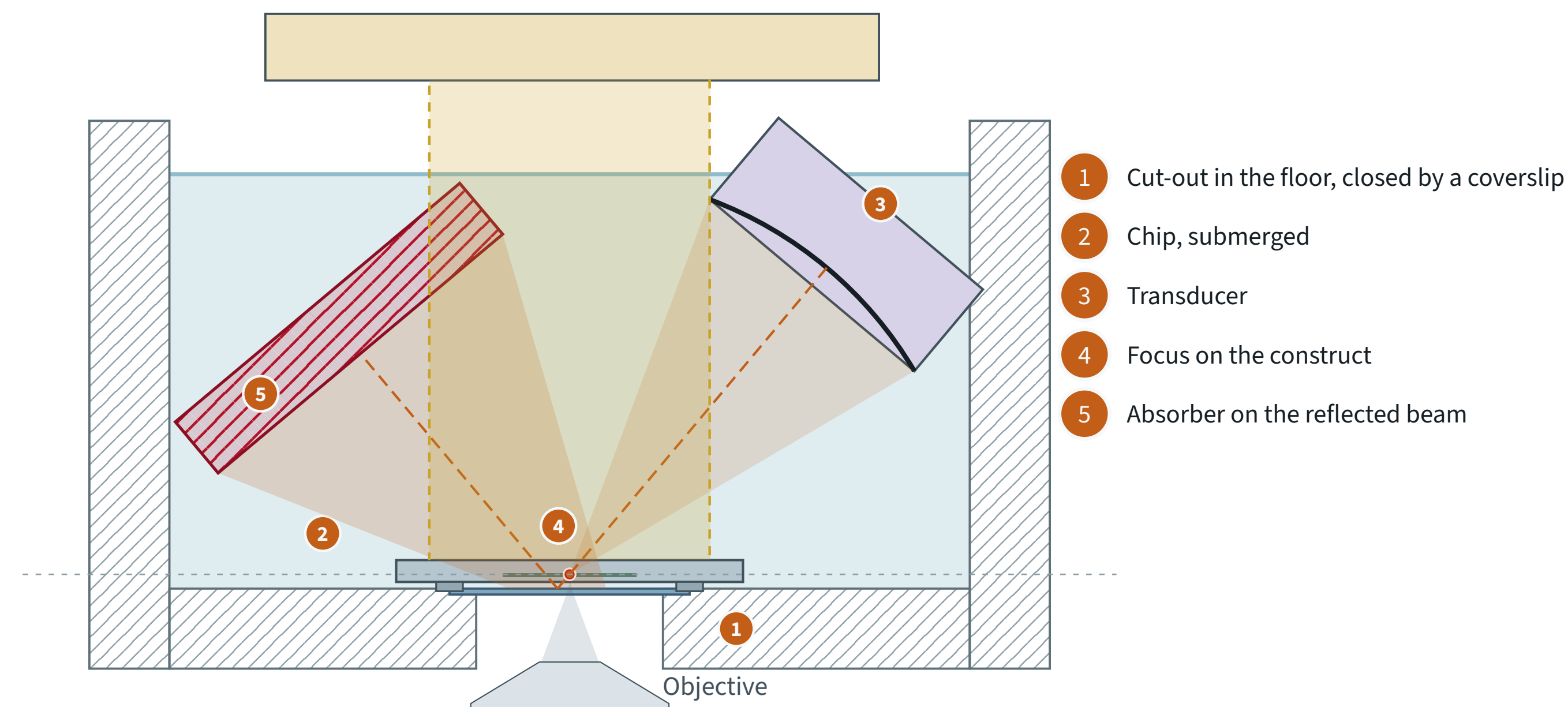
The culture route, inherited from earlier work in MimicLab at PoliMI.

Human cells, in three dimensions. Neurons and astrocytes in a gel construct, not a monolayer on glass. To verify (v) and (vi) we need the glia.

Concurrent stimulation and readout. Ultrasound, electrophysiology and calcium imaging on the same construct, during exposure rather than after.

Sub-MHz, transcranial-compatible. Staying below 1 MHz is what allows a result on this chip to be read against the in-vivo LIFU literature.

The inverted rig, 500 kHz, 40° tilt



Our current iteration of the rig design. It's a balancing act between optical path, acoustic transmission, thermal budgeting and practicality.

WHAT HAPPENS NEXT

Measure the field we deliver. Hydrophone scan plus a finite-element model of the real chip stack, with a thermal readout in the rig.

Settle on an experimental design. Which hypotheses to separate first, at which parameters, with which readouts. The model proposes the regimes where the six disagree and the platform will run them.

Close the loop. First round at the bench under concurrent electrophysiology and calcium imaging, then the measurements go back into the model.

REFERENCES

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